# Causal state-dependent local projections

Identification, estimation and inference for policy analysis

Weining Wang and Xiyu Jiao
DG ECFIN short course
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Based on David, Giacomini, Jiao and Wang (8 November 2026)

# Syllabus and timing

| | |
|---|---|
| 13:30–13:45 | Welcome and introduction |
| 13:45–14:30 | Identification and causal responses |
| 14:30–15:15 | Causal state-dependent LPs |
| 15:15–15:30 | Break |
| 15:30–16:15 | Estimation and inference |
| 16:15–16:45 | Practical R application |
| 16:45–17:00 | Discussion and Q&A |

Annotated reading list: six core readings and optional references by topic.

# Policy questions for this course

- Monetary policy and firm investment: how does responsiveness vary with financial health?
- A synthetic panel with a known response: how do specification and inference change the answer?
- Your own policy application: what is the shock, initial state and causal comparison?

The empirical example uses the current paper. The R practical uses simulated data with no empirical calibration.

# A local projection is a regression at each horizon

- Regress the future outcome on today's shock and relevant information
- Repeat for each horizon and plot the shock coefficients
- Choose levels, changes or cumulative outcomes to match the question
- Causal interpretation comes from the identification design

$$Y_{t+h} = \alpha_h + \beta_h X_t + \Gamma_h^\top W_{t-1} + u_{t+h}.$$

# Outcome definitions change the estimand

- Level response: the effect on $Y_{t+h}$
- Change response: the effect on $Y_{t+h} - Y_{t-1}$
- Cumulative flow: the effect on $\sum_{j=0}^{h} Y_{t+j}$
- A cumulative change in a level differs from a sum of flows
- Report whether a shock is one unit, one standard deviation or one percentage point

# LP and VAR estimation share the identification problem

- LP estimates each horizon directly. VAR propagates estimated dynamics
- With unrestricted lags they can recover the same population responses
- Finite-sample bias and precision depend on lag choices and restrictions
- Replacing a VAR with an LP does not make a shock exogenous

Sources: Jordà (2005), Plagborg-Møller and Wolf (2021).

# Session 1: Identification and causal responses

13:45–14:30

Target, assumptions, shock moments and failures of interpretation.

# Four common routes to identifying an LP

| Design | Economic identifying argument |
|---|---|
| Conditional exogeneity | Controls remove dependence between treatment and potential outcomes. |
| Recursive timing | Within-period exclusions isolate a structural innovation. |
| Observed shock | Narrative or high-frequency variation measures the intended exogenous shock. |
| External instrument | A relevant proxy is excluded from other determinants of the outcome. |

These routes can overlap. The LP estimates the response; the design gives it causal meaning.

Reading: Jordà and Taylor (2025); Stock and Watson (2018).

# Conditional exogeneity: what controls must accomplish

For each treatment value $d$, require consistency, support and

$$E[Y_{t+h}(d) \mid D_t, W] = E[Y_{t+h}(d) \mid W].$$

With an affine constant response, $E[Y_{t+h}(d) \mid W] = m_h(W) + \theta_h d$. Let $V_t = D_t - E[D_t \mid W]$. Then

$$\theta_h = \frac{E[V_t Y_{t+h}]}{E[V_t^2]}, \qquad E[V_t^2] > 0.$$

- Exogeneity connects observed and potential outcomes.
- Correct adjustment removes the nuisance $m_h(W)$.
- A finite set of linear lag controls need not implement this adjustment.

# Recursive timing is an identifying restriction

For reduced-form innovations ordered as outcome, then policy,

$$u_{y,t} = b_{11}\varepsilon_{1t}, \qquad u_{d,t} = b_{21}\varepsilon_{1t} + b_{22}\varepsilon_{2t}.$$

- The target policy shock has no contemporaneous effect on the first variable.
- Residualising $u_{d,t}$ on $u_{y,t}$ isolates $b_{22}\varepsilon_{2t}$ under this model.
- Within-quarter exclusions can be much stronger than within-minute exclusions.
- The resulting shock can enter an LP; the identifying assumption remains the timing restriction.

Reading: Plagborg-Møller and Wolf (2021).

# Observed shocks: measurement and economic content

- Narrative measures use policy records and the decision-maker's information.
- High-frequency surprises narrow the event window but may mix policy and other news.
- Predictability from pre-announcement data motivates careful information adjustment.
- A noisy proxy used directly need not have the same scale as the structural shock.

If $Y = \theta\varepsilon + u$ and $X = \varepsilon + \nu$ with mutually uncorrelated components,

$$\beta^{OLS} = \theta \frac{\mathrm{Var}(\varepsilon)}{\mathrm{Var}(\varepsilon) + \mathrm{Var}(\nu)}.$$

Examples: Romer and Romer (2004); Gertler and Karadi (2015); Bauer and Swanson (2023).

# LP-IV: relevance, exclusion and normalisation

With instrument $Q_t$, residualised instrument $\widetilde{Q}_t$, and $Y_{t+h} = \theta_h D_t + m_h(W) + u_{t+h}$,

$$E[\widetilde{Q}_t u_{t+h}] = 0, \quad E[\widetilde{Q}_t D_t] \neq 0 \quad \Longrightarrow \quad \theta_h = \frac{E[\widetilde{Q}_t Y_{t+h}]}{E[\widetilde{Q}_t D_t]}.$$

- Exclusion must hold for the horizon-$h$ error, including relevant dated shocks.
- A proxy design often identifies the response per unit impact movement in $D_t$.
- LP-IV can avoid invertibility under strong instrument conditions; this is not unconditional.
- Weak instruments require appropriate confidence sets; a valid first stage does not prove exclusion.

Reading: Stock and Watson (2018).

# Literature map: from an LP coefficient to our target

- **Estimation:** Jordà (2005); LP–VAR equivalence in Plagborg-Møller and Wolf (2021).
- **Causal objects:** Rambachan and Shephard (2025 revision).
- **Nonlinear responses:** Kolesár and Plagborg-Møller (2025): causal weighted marginal effects under suitable observed-shock or proxy assumptions.
- **State dependence:** Gonçalves et al. (2024): finite versus infinitesimal responses depend on the state mechanism.
- **Our course paper:** explicit conditions for $\mathrm{IRF}_h(z;\delta) = \delta g_h(z)$ with an unrestricted past-information nuisance.

**Next:** separate causal exogeneity, innovation orthogonality and an affine response in the shock.

# Potential outcomes: the intervention and the population

Let $Y_{i,t+h}(x)$ be the outcome when the date-$t$ shock is set to $x$.

$$\mathrm{IRF}_h(z;\delta) = E[Y_{i,t+h}(X_t + \delta) - Y_{i,t+h}(X_t) \mid Z_{i,t-1} = z].$$

- Shift the realised innovation by $\delta$ and average within the initial-state group.
- Later outcomes and states evolve under the intervention.
- Consistency: the observed outcome equals $Y_{i,t+h}(X_t)$.
- Both $X_t$ and $X_t + \delta$ must lie in the relevant conditional support.

Source: manuscript, Section 3.1. A fixed comparison of $x$ and $x + \delta$ gives the same slope under the affine condition below.

# The causal comparison

- Specify outcome, horizon, shock units and initial state
- Change the date-t innovation; let later outcomes evolve
- Condition on the initial state and allow subsequent states to evolve

$$\text{IRF}_h(z;\delta) = \delta g_h(z), \qquad \Delta_h = g_h(z_A) - g_h(z_B)$$

# Get the timing right

- State and controls: information available before the shock
- Shock: date t; outcome: date t+h
- Respect publication dates, anticipation and data revisions

# From causal assumptions to a response function

For every admissible $x$, causal mean exogeneity requires

$$E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}, X_t] = E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}].$$

Consistency and the observed conditional-mean restriction imply

$$E[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}] = r_{i,h}(\mathcal{F}_{t-1}) + g_h(Z_{i,t-1})x.$$

Subtract the counterfactual means and average over initial-state groups:

$$\mathrm{IRF}_h(z;\delta) = \delta g_h(z).$$

## Roles of the assumptions

Exogeneity connects the observed and counterfactual means. The affine restriction makes the slope independent of shock size and starting value.

# The identification conditions

- Causal exogeneity links shocks to potential outcomes
- An unpredictable shock removes the past-information nuisance
- The conditional mean is affine in the shock
- The state captures the relevant response coefficient; support is positive

$$E[X_t \mid \mathcal{F}_{t-1}] = 0, \qquad E[Y_{i,t+h} \mid \mathcal{F}_{t-1}, X_t] = g_h(Z_{i,t-1})X_t + r_{i,h}(\mathcal{F}_{t-1})$$

# The shock moment: a derivation

Write $Y = g_h(Z)X + r(\mathcal{F}) + u$, where $E[u \mid \mathcal{F}, X] = 0$, $Z$ is $\mathcal{F}$-measurable and $E[X \mid \mathcal{F}] = 0$.

$$
\begin{aligned}
E[XY \mid Z] &= g_h(Z)E[X^2 \mid Z] + E[Xr(\mathcal{F}) \mid Z] + E[Xu \mid Z], \\
E[Xr(\mathcal{F}) \mid Z] &= E[r(\mathcal{F})E[X \mid \mathcal{F}] \mid Z] = 0, \\
E[Xu \mid Z] &= E[XE[u \mid \mathcal{F}, X] \mid Z] = 0.
\end{aligned}
$$

If $E[X^2 \mid Z = z] > 0$, the ratio of moments identifies $g_h(z)$. Statistical identification alone does not establish the counterfactual interpretation.

# Why the past-information nuisance can be left unrestricted

Under A1–A3, for any predetermined basis $\phi_J(Z)$ and controls $W$,

$$E[X_t \phi_J(Z_{i,t-1}) r_{i,h}(\mathcal{F}_{t-1})] = 0, \qquad E[X_t \phi_J(Z_{i,t-1}) W_{i,t-1}^\top] = 0.$$

- The m.d.s. condition removes arbitrary nonlinear functions of the past from the shock moment.
- Predetermined controls can improve precision without spanning the entire nuisance.
- A shock residual orthogonal to a few selected lags need not satisfy $E[X_t \mid \mathcal{F}_{t-1}] = 0$.
- Sample transformations, including within-unit demeaning, require a separate justification.

Source: manuscript, Section 3.1, especially footnote 5.

# Shock construction is an identification decision

- A policy change need not be an exogenous policy shock
- Assess anticipation, information effects and omitted news
- Residualizing a series does not establish causal exogeneity
- An instrument requires an IV design and IV inference

# Serial correlation creates a nuisance term

Suppose $X_t = \rho_x X_{t-1} + \eta_t$ and

$$Y_{t+h} = g_h X_t + \gamma_h X_{t-1} + u_{t+h}, \qquad E[X_t u_{t+h}] = 0.$$

For stationary, mean-zero $X_t$, a regression omitting $X_{t-1}$ has slope

$$\frac{E[X_t Y_{t+h}]}{E[X_t^2]} = g_h + \gamma_h \rho_x.$$

- Serial correlation creates bias here through the omitted lag.
- Correct lag controls or correctly specified residualisation can remove this statistical term.
- Neither operation proves that $\eta_t$ is a causally exogenous innovation.

# A nonlinear shock response changes the estimand

Consider $Y = g(z)X + c(z)X^3 + u$, with $X \sim N(0,1)$ independent of the state and causally exogenous.

$$\beta^{LP}(z) = \frac{E[XY \mid Z = z]}{E[X^2 \mid Z = z]} = g(z) + 3c(z).$$

The paper's finite-shift target is

$$E[Y(X+\delta) - Y(X) \mid Z = z] = \delta\{g(z) + 3c(z)\} + c(z)\delta^3.$$

- Multiplying the LP slope by $\delta$ misses $c(z)\delta^3$.
- For a fixed comparison from 0 to $\delta$, the response is $g(z)\delta + c(z)\delta^3$.
- Flexible dependence on $z$ does not remove nonlinearity in $X$.

# Controls and fixed effects need a purpose

- Use pre-shock information for the intended total response
- Post-shock controls can remove transmission channels
- Date fixed effects absorb the common shock main effect
- Unit fixed effects bring their own dynamic-panel issues

# Discuss your identification design

- Name your outcome, shock, initial state and horizon
- Identify one plausible threat to causal interpretation
- Decide whether the current aim is descriptive or causal

# Session 2: Causal state-dependent local projections

14:30–15:15

Propagation, endogenous initial states and functional-form restrictions.

# Worked propagation example

Assume $Y_{it} = \rho Y_{i,t-1} + a(Z_{i,t-1})X_t + \varepsilon_{it}$, and future innovation terms have zero conditional mean under the intervention.

$$
\begin{aligned}
E[Y_{it}(x+\delta) - Y_{it}(x) \mid \mathcal{F}_{t-1}] &= a(Z_{i,t-1})\delta, \\
E[Y_{i,t+1}(x+\delta) - Y_{i,t+1}(x) \mid \mathcal{F}_{t-1}] &= \rho a(Z_{i,t-1})\delta, \\
g_h(z) &= \rho^h a(z).
\end{aligned}
$$

With $\rho = .6$ and $a(z) = 1 - z^2$, $g_0(0) = 1$, $g_0(.75) = .4375$, and their impact contrast is .5625.

## Scope

Stable linear propagation and the future-innovation restrictions are substantive assumptions.

# An endogenous state can still be admissible

- Initial conditions may depend on earlier shocks
- Stable propagation preserves linearity in the current shock
- Changing propagation or shock nonlinearities need more care

$$Y_{it} = a(Z_{i,t-1})X_t + \rho Y_{i,t-1} + \varepsilon_{it} \quad \Longrightarrow \quad g_h(z) = \rho^h a(z)$$

# When state transitions alter propagation

If $Y_{t+1} = \rho(Z_t)Y_t + \varepsilon_{t+1}$ and the date-$t$ intervention changes $Z_t$, then

$$Y_{t+1}(x+\delta) - Y_{t+1}(x) = \rho(Z_t(x+\delta))Y_t(x+\delta) - \rho(Z_t(x))Y_t(x).$$

- A single constant propagation coefficient generally cannot factor out.
- The response can depend on the starting shock value and the shock size.
- Conditioning on realised $Z_t$ can condition on a consequence of treatment.
- Initial-state heterogeneity and interventions on the state are distinct causal questions.

# Three DGPs and the scope of the method

| Dynamic structure | What the flexible LP can recover |
|---|---|
| Linear shock exposure, fixed propagation | $g_h(z)$ under A1–A3 and support conditions |
| Shock-dependent propagation | Generally a projection; the affine representation can fail |
| Nonlinear shock exposure | Generally a projection; the finite-shift response depends on $X_t$ and $\delta$ |

A

fixed state alone does not establish A3. First-order HA/HANK solutions can satisfy A3 within their linear approximation. A nonlinear economy requires checking the approximation over the relevant shock range.

Source: manuscript, Sections 2 and 3.2. Scalar average-effect interpretations have their own assumptions and do not recover the full state function.

# Coarsening a state changes the weights

Within a reported state group, two equally common latent types have slopes 1 and 3, and shock variances 1 and 4. Assume conditional shock means are zero.

$$\text{Average causal slope} = \tfrac{1}{2}(1+3) = 2, \qquad \text{LP slope} = \frac{\frac{1}{2}(1)(1) + \frac{1}{2}(4)(3)}{\frac{1}{2}(1) + \frac{1}{2}(4)} = 2.6.$$

More generally, with latent slope $a$ and variance $v$,

$$\beta^{LP}(z) - E[a \mid z] = \frac{\mathrm{Cov}(v, a \mid z)}{E[v \mid z]}.$$

A coarse recession/expansion classification can therefore change the target, even when every latent-type response is causal.

# The linear interaction is a benchmark

- Estimate the common shock and its interaction with the state
- A small interaction does not establish homogeneous effects
- Compare the fitted function with a flexible alternative

$$Y_{i,t+h} = \alpha_h X_t + \beta_h X_t Z_{i,t-1} + W_{i,t-1}^{\top} \gamma_h + u_{i,t+h}$$

# A linear interaction averages slopes

Under the assumptions of Proposition 2,

$$\beta_h^{\mathrm{linearSD}} = \int_{\mathcal{Z}} \omega_h(z) g_h'(z)\, dz, \qquad \omega_h(z) \geq 0, \quad \int_{\mathcal{Z}} \omega_h(z)\, dz = 1.$$

- This coefficient summarizes local slopes of the response curve.
- A positive coefficient does not establish that the curve rises everywhere.
- Opposing slopes can cancel even when state contrasts are large.
- Recovering a response function requires estimating its shape.

Source: manuscript, Proposition 2 and Appendix A. Conditional homoskedasticity gives $\beta_h = \mathrm{Cov}(Z, g_h(Z))/\mathrm{Var}(Z)$.

# Why a zero linear interaction can hide heterogeneity

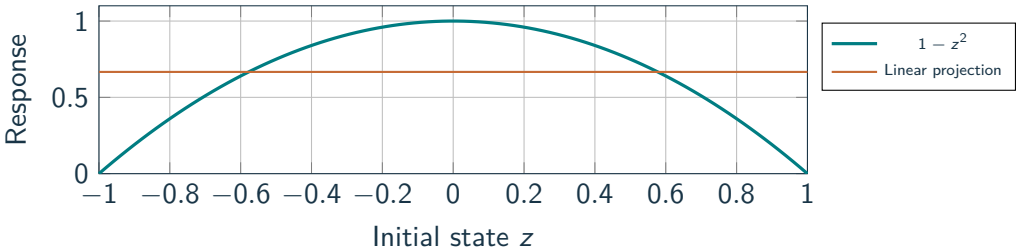
Let $Z \sim U[-1, 1]$, with an independent mean-zero shock of constant variance, and $g(z) = 1 - z^2$. Project $g(Z)$ onto $(1, Z)$:

$$\beta = \frac{\mathrm{Cov}(Z, 1 - Z^2)}{\mathrm{Var}(Z)} = 0, \qquad \alpha = E[1 - Z^2] = \frac{2}{3}.$$

- Symmetry cancels the positive and negative slopes.
- The response is 1 at the centre and 0 at the boundaries.
- Testing one linear interaction does not test general homogeneity.
- The result uses these specific distribution and weighting assumptions.

# Why the linear interaction can miss heterogeneity

- A symmetric hump can have an almost zero linear slope
- A zero interaction coefficient need not imply equal responses
- This figure shows a population illustration, not ECFIN estimates



# Common shocks and response normalization

With unrestricted date effects,

$$Y_{i,t+h} = \alpha_{t,h} + X_t g_h(Z_{i,t-1}) + \cdots .$$

For any constant $c$, the same fitted values follow from

$$g_h^*(z) = g_h(z) + c, \qquad \alpha_{t,h}^* = \alpha_{t,h} - cX_t.$$

- The common level needs a normalization or additional identifying structure.
- A contrast $g_h(z_A) - g_h(z_B)$ is invariant to that level shift.
- A constant is already in the span of the full spline basis. Adding a separate $X_t$ column causes collinearity.

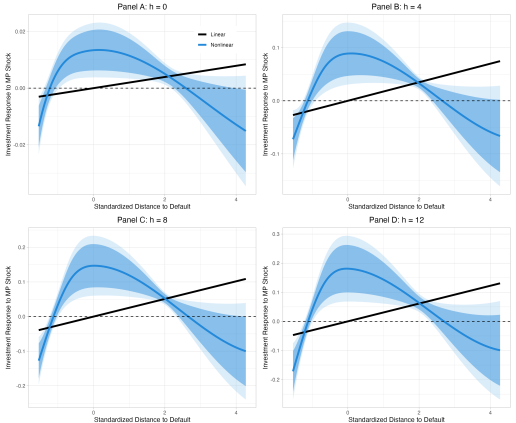
# Application: monetary policy and firm investment

$$k_{i,t+h} - k_{i,t-1} = \alpha_{sth} + g_h(Z_{i,t-1})X_t + \rho_h \Delta k_{i,t-1} + W_{i,t-1}^{\top}\gamma_h + u_{i,t+h}.$$

- Quarterly Compustat capital stock and high-frequency monetary surprises.
- Initial state: standardized distance to default (D2D). Higher values mean safer firms.
- Figures show cumulative investment responses to a 25 b.p. surprise rate cut.
- Sector-time-horizon effects absorb the common response. Interpret the plotted curves under the paper's level normalization.

Source: manuscript, Section 6, equation (27). This is separate from the synthetic R exercise.

# Firm responses across the financial-state distribution




Source: manuscript, Figure 3. Differential cumulative responses to a 25 b.p. cut at horizons 0, 4, 8 and 12 quarters. Dark/light shading: 68%/90% state-uniform bands, separately at each horizon.

# Reading the empirical response curve

- The flexible estimate peaks around mean D2D and falls toward both tails.
- Responsiveness increases as firms become safer in the left tail, then decreases to the right of the peak.
- A linear interaction imposes a single slope across both regions.
- Relative to the earlier empirical specification, controls and fixed-effect treatment also differ. The full change is not attributable solely to the spline.

Discussion: which pair of states would you compare, and what covariance would you need?

Source: manuscript, Section 6.1. A response pattern does not uniquely identify its economic mechanism.

# Session 2 discussion

- Which variable is the initial state in your project?
- Could it be affected by the current intervention?
- Would coarsening the state alter the weighting?
- Which restriction makes the response affine in the shock?

# Break

15:15–15:30

# Session 3: Estimation and inference

15:30–16:15
Weining Wang and Xiyu Jiao

# A flexible response with cubic B-splines

- Interact every basis column with the shock
- Keep the same basis when predicting at new state values
- Choose J from 4 to 20 using an AIC criterion

$$\widehat{g}_h(z) = \phi_J(z)^\top \widehat{b}_h, \qquad D_{it} = (X_t \phi_J(Z_{i,t-1})^\top, W_{i,t-1}^\top)^\top$$

```
B <- splines::bs(z, degree=3, knots=knots,
                 intercept=TRUE)
D <- cbind(B*x, W)
```

# Spline estimation as a partially residualised regression

For a fixed horizon, stack $Q_{it} = X_t \phi_J(Z_{i,t-1})$ and controls $W_{it}$ into matrices $Q, W$.

$$M_W = I - W(W'W)^{-1}W', \qquad \widehat{b}_h = (Q' M_W Q)^{-1} Q' M_W Y_h.$$

$$\widehat{g}_h(z) = \phi_J(z)' \widehat{b}_h.$$

- Full column rank is necessary. Use QR methods numerically.
- Under A1–A3, m.d.s. orthogonality removes the past-information nuisance. Controls may improve precision. If these conditions fail, adding selected controls is not an automatic repair.
- Retain the training knots and boundary knots for prediction.

# Basis dimension: fit, variance and bias

The teaching implementation compares $J = 4, \ldots, 20$ using

$$\mathrm{AIC}(J) = n_h \log(\mathrm{RSS}_J / n_h) + 2p_J,$$

where $p_J$ counts all fitted coefficients.

- A larger basis can reduce approximation error and increase variance.
- This criterion is a teaching selection rule, not a dependence-adjusted optimality theorem.
- For inference about the true function, approximation bias must be negligible or explicitly corrected.
- Re-estimate with a few fixed dimensions and report sensitivity.

# The dependence structure determines standard errors

- Time series: assess dependence of regression scores across dates
- Country panels: assess within-country and cross-country dependence
- Common shocks: many firms do not create many independent shock dates
- Lag-augmented time-series LPs can permit heteroskedasticity-robust inference under their assumptions

Source: Montiel Olea and Plagborg-Møller (2021). Their result does not automatically extend to this flexible panel estimator.

# Why precision can change across the state space

The paper lets the effective-score dependence scale grow with $N$:

$$\chi_{h,J,N} = 1 \vee \lambda_{\max}(\Omega_{h,J,N}), \qquad \|\widehat{b}_h - b_h\|_2 = O_p(J^{1/2-\kappa}) + O_p\left(\sqrt{\frac{J\chi_{h,J,N}}{NT}}\right).$$

- Weak score dependence: stochastic coefficient error of order $\sqrt{J/(NT)}$.
- Strong aggregate components, $\chi \asymp N$: order $\sqrt{J/T}$.
- Aggregate components can load differently at different states. One common rate need not describe every point on the curve.
- Use the estimated state-specific standard error when studentizing.

Source: manuscript, Theorem 1. Joint growth, moment, approximation and dependence restrictions apply. This is a coefficient rate, not a universal pointwise rate.

# Overlapping outcomes induce serial dependence

For $Y_t = \rho Y_{t-1} + \epsilon_t$, the $h$-step forecast error contains

$$e_{t,h} = \sum_{j=1}^{h} \rho^{h-j} \epsilon_{t+j}.$$

Adjacent forecast errors share future innovations. In a panel, common components can also correlate observations within a date.

- Treating all unit-date rows as independent can understate uncertainty.
- Clustering only within dates does not allow serial dependence between dates.
- Date-aggregated HAC addresses both in the teaching design, under suitable time-series regularity.

# HAC covariance: the full construction

With full regressor vector $D_{it}$ and residual $\widehat{u}_{i,t+h}$, form date sums $s_t = \sum_i D_{it}\widehat{u}_{i,t+h}$.

$$\widehat{\Omega} = \sum_t s_t s_t' + \sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\sum_{t=\ell+1}^{T_h}(s_t s_{t-\ell}' + s_{t-\ell}s_t').$$

$$\widehat{V} = (D'D)^{-1}\widehat{\Omega}(D'D)^{-1}.$$

- These are sums throughout; no additional division by $n_h$ belongs in this formula.
- Extract the spline coefficient block for function inference.
- Order scores by calendar date and assess bandwidth sensitivity.

# The inference question determines the uncertainty measure

- One preselected horizon and state: a pointwise interval
- Two specified states: an interval for their difference
- A whole curve: a simultaneous band over the stated domain
- Significant in one group and insignificant in another does not establish a difference

# A contrast needs the covariance between fitted responses

For $\Delta = g(z_A) - g(z_B)$, set $a = \phi(z_A) - \phi(z_B)$.

$$\widehat{\Delta} = a'\widehat{b}, \quad \widehat{\mathrm{Var}}(\widehat{\Delta}) = a'\widehat{V}_b a.$$

Equivalently,

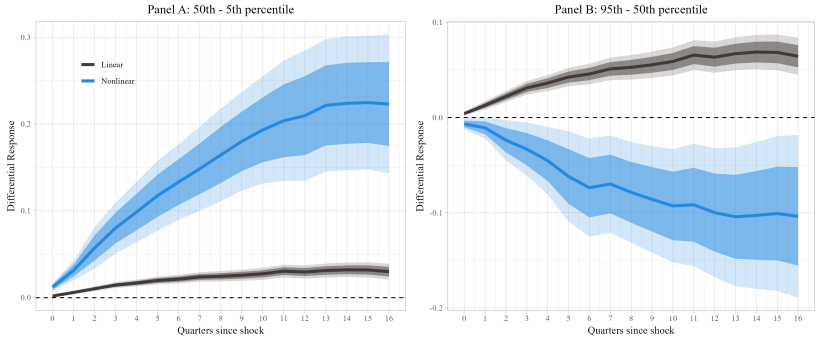
$$\mathrm{Var}(\widehat{g}_A - \widehat{g}_B) = V_A + V_B - 2C_{AB}.$$

If both standard errors are .10 and the covariance is .008, the contrast standard error is $\sqrt{.02 - .016} \approx .063$.

## Interpretation

Overlapping individual confidence intervals do not determine whether a contrast is significant.

# Empirical contrasts answer a specific question




Left: median minus 5th-percentile D2D. Right: 95th percentile minus median.

Source: manuscript, Figure 4. Responses to a 25 b.p. cut. Shading gives 68% and 90% pointwise intervals for each prespecified state contrast at each horizon, not a simultaneous band across horizons.

# Pointwise intervals and simultaneous bands

- Pointwise: one preselected state at one horizon
- Simultaneous band: the whole specified state grid at one horizon
- Studentize Gaussian draws by each state's standard error
- A state band is not a band over every horizon

$$c_{.95} = q_{.95}\left[\max_{z \in \mathcal{Z}_G} \frac{|\phi(z)^\top \widehat{V}_b^{1/2} \xi|}{\widehat{\sigma}(z)}\right], \qquad \widehat{g}(z) \pm c_{.95}\widehat{\sigma}(z)$$

```
curve <- infer_curve(impact,
  grid=seq(-.8,.8,length.out=161))
```

# Gaussian simulation of a simultaneous state band

At one fixed horizon and a prespecified grid $\mathcal{Z}_G$:

1. Compute $\widehat{\sigma}(z)^2 = \phi(z)' \widehat{V}_b \phi(z)$.
2. Draw $\xi^{(b)} \sim N(0, I_J)$ and set $G^{(b)}(z) = \phi(z)' \widehat{V}_b^{1/2} \xi^{(b)}$.
3. Record $M^{(b)} = \max_{z \in \mathcal{Z}_G} |G^{(b)}(z)| / \widehat{\sigma}(z)$.
4. Use the .95 quantile of these maxima as $c_{.95}$.

$$\widehat{g}(z) \pm c_{.95} \widehat{\sigma}(z).$$

A finite grid approximates a continuum. Coverage requires covariance consistency, distributional approximation and adequate bias control.

# A confidence band also needs bias control

- Spline approximation error must be small relative to sampling uncertainty
- AIC selects a fit. It does not by itself guarantee confidence-band coverage
- Inspect sensitivity to basis size, support and the state range
- The practical illustrates asymptotic inference, not a finite-sample coverage guarantee

# Simulation evidence: narrow bands can still undercover

Paper simulation, $N = 500$, $T = 200$, $h = 4$, AIC selection, nominal 95% state-uniform bands:

| Specification | Coverage | Average width |
|---|---|---|
| Without intermediate shock terms | 0.89 | 1.30 |
| With intermediate shock terms | 0.83 | 0.05 |

- Extra shock interactions can remove common error variation in this DGP.
- The large precision gain does not deliver nominal coverage here.
- The paper retains the unaugmented baseline because additional terms increase dimension and can be unstable.

Source: manuscript, Table 3. These are Monte Carlo results, separate from the course simulation.

# Diagnostics before interpretation

- Count independent shock dates, not only panel rows
- Inspect support, shock variation and influential dates
- Check missingness, sample composition and long horizons
- Inspect sensitivity to state definition and spline complexity

# Session 4: Practical R application

16:15–16:45
Work through the participant handout.

# The practical: a 30-minute route

| Minutes | Participant task |
|---|---|
| 0–7 | Inspect timing, keys and state support |
| 7–15 | Estimate the impact response and plot it |
| 15–23 | Contrast two states and compare uncertainty |
| 23–30 | Sensitivity and a short interpretation |

Use the participant handout and `exercises/participant_template.R`. The setup file generates synthetic data and defines the functions without running the full horizon loop. Optional extensions follow after the core tasks.

# The simulated panel

- 100 units observed over 240 shock dates
- A common shock; a pre-shock financial-state index
- Known response: a hump in the state, with stable decay
- Synthetic units: no empirical policy calibration

$$g_h(z) = 0.6^h(1 - z^2)$$

# Setup and files

```
source("code/R/exercise_setup.R")
impact <- estimate_horizon(0, Jset=4:10)
```

Run from the extracted course folder. All data in this exercise are simulated.

# R lab: check timing and the sample

- Check unique unit-date keys and missing periods
- Verify that z is pre-shock and x is common within date
- Inspect state support and the number of shock dates

```
stopifnot(!anyDuplicated(d[c("id","t")]))
quantile(d$z, c(.05,.5,.95))
length(unique(d$t))
```

# R lab: estimate the response function

- Run the simulated example and fit the impact response
- Inspect the AIC table and selected basis dimension
- Plot the response only over the chosen data-supported range

```
impact <- estimate_horizon(0, Jset=4:10)
impact$AIC
impact$J
infer_contrast(impact)
```

# R exercise: the linear benchmark

- Compare the impact response under a linear interaction and a spline
- The true impact function is $1 - z^2$
- Use the benchmark to diagnose functional-form restrictions

```
m <- lm(y ~ x + I(x*z) + ylag, data=d)
zgrid <- seq(-.8,.8,length.out=161)
glin <- coef(m)["x"] + coef(m)["I(x * z)"]*zgrid
lines(zgrid, glin, col="orange")
```

# R lab: inference for a state contrast

- Choose the states before looking for a large difference
- Use the covariance between the two fitted responses
- Report the estimate, interval, horizon and shock units

$$a = \phi(z_A) - \phi(z_B), \qquad \widehat{\Delta} = a^\top \widehat{b}, \qquad \mathrm{se}(\widehat{\Delta}) = \sqrt{a^\top \widehat{V}_b a}$$

```
infer_contrast(impact, zA=0, zB=.75)
```

# An interpretation with the right scope

## Reporting template

At horizon ____, the estimated response per unit shock at state ____ is _____. The interval is _____ and covers a _____ claim.

- State the shock normalization and outcome units.
- Report the state contrast directly, with its own uncertainty.
- In this practical, describe a synthetic demonstration.
- For real data, add the identification argument and the main threat to it.

# Optional R extension: an endogenous initial state

Keep the same outcome equation, but let

$$Z_{it} = \tanh(0.75 Z_{i,t-1} + 0.25 X_t + 0.3 v_{it}).$$

- The date-$t$ shock changes the next state. The regressor still uses $Z_{i,t-1}$.
- Independent future shocks and fixed outcome propagation preserve $g_h(z) = 0.6^h(1 - z^2)$.
- This exercise tests the distinction between endogenous evolution and post-shock conditioning.

```
endog <- simulate_panel(state="endogenous")
efit <- estimate_horizon(0, Jset=4:10, data=endog)
infer_contrast(efit)
```

# Discussion and Q&A

16:45–17:00

Applications to DG ECFIN policy analysis and participants' research questions.

# Before using an LP result for policy

- Can I defend the shock and initial-state timing?
- Does the response-function interpretation fit this setting?
- Does inference match dependence and the scope of the claim?
- Have I reported support, units and meaningful sensitivity?

# Open questions and research clinic

- Describe your data, outcome, shock and initial state
- What would make the result useful for your project?
- Identify the main identification or inference difficulty
- Agree one concrete next analysis or diagnostic

# Optional extensions

Additional derivations and diagnostics for follow-up reading.

# Horizon construction must respect the calendar

- A lead of four means four periods, even when observations are missing
- Match unit and date keys instead of shifting rows blindly
- Report observations and shock dates separately at each horizon
- A common sample helps assess composition sensitivity

```
key <- paste(d$id, d$t, sep=":")
h <- 4
yh <- d$y[match(paste(d$id,d$t+h,sep=":"), key)]
sum(!is.na(yh))
```

# Responses across horizons

- Rebuild the outcome and estimation sample at each horizon
- Distinguish levels, changes and cumulative outcomes
- Pointwise horizon intervals do not cover the whole path jointly

```
fits <- lapply(0:8, estimate_horizon)
```

# Replicate first, then assess sensitivity

- Match the published target, sample, transformations and units
- Match controls, horizons, tuning and covariance conventions
- Reproduce a reported number before changing the design
- Then test sensitivity to choices that matter for the claim

```
estimate_horizon(4, Jset=7)
estimate_horizon(4, bandwidth=8)
```

# Appendix: weighted and unweighted state targets

- The ordinary subpopulation response averages causal slopes
- The LP additionally weights by conditional shock variance
- Their difference is a conditional covariance divided by mean variance
- Constant conditional variance is sufficient for equality

$$\frac{\mathrm{IRF}_h(z;\delta)}{\delta} = E[\widetilde{g}_h(s) \mid Z = z],$$

$$g_h^{\mathrm{LP}}(z) - E[\widetilde{g}_h(s) \mid Z = z] = \frac{\mathrm{Cov}(v(s,z), \widetilde{g}_h(s) \mid Z = z)}{E[v(s,z) \mid Z = z]}, \quad v(s,z) = E[X^2 \mid s, Z = z].$$

# Appendix: finite shifts with nonlinear shock exposure

For $f_z(x) = g(z)x + c(z)x^3$ and $X \sim N(0,1)$,

$$E[f_z(X+\delta) - f_z(X)] = g(z)\delta + c(z)\{3\delta E[X^2] + 3\delta^2 E[X] + \delta^3\}.$$

Thus the average derivative equals $g(z) + 3c(z)$, while the average finite shift also contains $c(z)\delta^3$.

```
nonlinear_shift_demo(z=0, delta=1, cubic=.25)
```

The output compares the LP slope times the shift with the finite-shift target. At these values the exact discrepancy is 0.25.

# Appendix: FWL and the limits of linear controls

Suppose $Y = \theta D + m(W) + u$, $E[u \mid D, W] = 0$, and $V^L = D - P_W D$, where $P_W$ is the linear projection including an intercept.

$$\beta^{OLS} = \theta + \frac{E[V^L m(W)]}{E[(V^L)^2]}.$$

- Correct conditional exogeneity does not guarantee a correct linear adjustment.
- The extra term vanishes if $m(W)$ is in the control span, or if $P_W D = E[D \mid W]$.
- Exact two-step FWL equivalence requires the same OLS sample and controls.
- Changing samples, controls or estimation methods can break that equivalence.

Further reading: Lloyd and Manuel (2024).

# Appendix: why a nonlinear LP can still be causal

For a causally exogenous continuous shock $X$ and smooth response $\Psi_h(x) = E[Y_{t+h}(x)]$,

$$\beta_h^{LP} = \int \omega(x)\Psi_h'(x)\,dx, \qquad \omega(x) = \frac{E[(X - EX)1\{X \geq x\}]}{\mathrm{Var}(X)}.$$

Under interval support and suitable integrability, $\omega(x) \geq 0$ and $\int \omega(x)dx = 1$.

- The coefficient averages marginal causal effects with specific weights.
- With Gaussian shocks and regularity, it equals the average derivative.
- It need not equal a derivative at a chosen point or a finite-shift effect.

Observed-shock illustration of Kolesár and Plagborg-Møller (2025). Proxy and conditional-control designs require additional assumptions.

# References and course sources


- David, J. M., R. Giacomini, X. Jiao and W. Wang (2026). *Causal State-Dependent Local Projections in Micro-Macro Data*. Manuscript dated 8 November 2026. Core source for assumptions, estimation, inference and firm results.
- Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1), 161–182.
- Plagborg-Møller, M. and C. K. Wolf (2021). Local projections and VARs estimate the same impulse responses. *Econometrica*, 89(2), 955–980.
- Montiel Olea, J. L. and M. Plagborg-Møller (2021). Local projection inference is simpler and more robust than you think. *Econometrica*, 89(4), 1789–1823.


Course code and synthetic exercises are teaching illustrations. They do not reproduce the empirical dataset or establish finite-sample coverage.

# References: identification and nonlinear responses


- Jordà and Taylor (2025). Local Projections. *JEL* 63(1), 59–110.
- Stock and Watson (2018). Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments. *Economic Journal* 128, 917–948.
- Rambachan and Shephard (2025 revision). When do common time series estimands have nonparametric causal meaning? arXiv:1903.01637v4.
- Kolesár and Plagborg-Møller (2025). Dynamic Causal Effects in a Nonlinear World: the Good, the Bad, and the Ugly. *JBES* 43(4), 737–754.
- Kitagawa, Wang and Xu (2025). Nonlinearity in Dynamic Causal Effects: Making the Bad into the Good, and the Good into the Great? *JBES* 43(4), 770–777.
- Gonçalves, Herrera, Kilian and Pesavento (2024). State-dependent local projections. *JoE* 244(2), 105702.


Reading list: priorities, annotations and links. Companion notes: assumptions and derivations.

# References: shock construction and controls


- Romer and Romer (2004). A New Measure of Monetary Shocks: Derivation and Implications. *AER* 94(4), 1055–1084.
- Gertler and Karadi (2015). Monetary Policy Surprises, Credit Costs, and Economic Activity. *AEJ: Macro* 7(1), 44–76.
- Bauer and Swanson (2023). A Reassessment of Monetary Policy Surprises and High-Frequency Identification. *NBER Macroeconomics Annual*.
- Lloyd and Manuel (2024). Controls, not shocks: estimating dynamic causal effects in macroeconomics. Bank of England Staff Working Paper 1079.


Compare how each design uses information and shock measures before copying its regression specification.